Element A

Problem Identification:
Problem Statement:
The Nashville Zoo faces a common problem that many animal-related organizations face -- storage and transportation of animals. For the safety of both the animals and the vets, the zoo wants a box to hold and transport the dragons for medical examinations. Zookeepers would like to perform ultrasounds and X-rays through the box without having to handle the dragon directly. The dragons are of varying sizes and weights, requiring a variable box design. The zookeepers must be careful when handling the komodo dragons as they are very strong and dangerous, with their tails whipping, sharp claws, and venomous bite. The zookeepers' current carrying system for the Komodo dragons is unsafe and provides low visibility of the dragons inside, as it is a large plastic tube (see included picture). The zookeepers wish to have a device that traps the dragon while preventing them from moving to facilitate the usage of medical probes, X-rays, and ultrasounds. That is, the zookeepers need a way to transport the Komodo dragons and run several medical tests without risking injury to the zookeepers and the dragons.

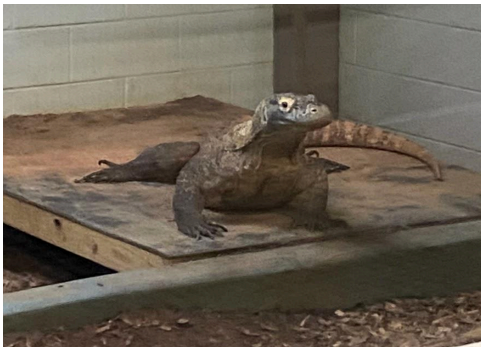
Through an interview with Nathaniel Hanna of the Nashville Zoo via Zoom, team members gained a better understanding of the problems that Zoo staff experience with their current solution. At the moment, Komodo dragons are transported into the medical examination bay using a rudimentary plastic tube. Hanna explained that, at the moment, handlers bait the dragons into the tube using frozen bait - often food and treats in the form of frozen rodents- before trapping them in the tube with lids on both sides. Hanna specifically mentioned the use of Y-shaped poles to push and prod the dragons during transportation as well as the use of riot shields during the operation or examination of the dragons. Hanna noted that the use of sedation is presently reserved only for removing the dragon from the box and is not currently used for ultrasound or capture of the dragons. These current protocols place the caretakers and the dragons alike in avoidable danger.
Concerns of primary stakeholders:

The primary stakeholders include the Nashville Zoo, Nathaniel Hannah (head of reptiles), zookeepers, and the Veterinarians at the Nashville Zoo. The main concerns of the zookeepers include the safety of the zookeepers and the safety of the dragons. They also need the solution to accommodate multiple sizes of Komodo Dragons, from 9 feet long and 200 lbs adult males to smaller 90 lbs females and even smaller juvenile males. The solution should be easily and quickly adaptable between these sizes. The veterinarian likewise needs the box to perform medical tests and procedures on the dragons, mainly consisting of drawing blood from their tails, performing x-rays, and taking ultrasounds. To this end, the Nashville Zoo Zookeepers specifically noted access to the dragon's torso and the ability to probe tails as necessary facets of standard medical procedure.
Any solution needs to be easy to clean out between uses, ideally waterproof so that it can be hosed out. They mentioned that it would be great to have a way to trim the dragon's claws in the box, however, that may cause more of a problem than it would solve.

Simultaneously, the challenges mentioned above posed by the venomousness of Komodo Dragons pose challenges to safely performing these procedures.

- Accommodations to poke the tail
- 6"x18" opening to perform tasks
- Ability to scan with X-ray

Komodo dragon size
The dragons range in size, with young males and females around 4 feet long and around 80- 90 lbs, up to large males at 9-10 feet long and 200 lbs. These are some photos of the dragons that they have at the zoo. On the left is their largest male dragon, around 9 feet long and 200 lbs. The two on the right are females that the zoo has, each around 6 feet long and 90 lbs. In addition, the zoo has two juvenile males smaller than the females and two hatchlings from the past summer.



The boxes must be able to adjust to the different sizes of the dragons currently and as they grow.